P122 | Advanced Topics in Plasma Physics

Shocks and Particle Acceleration

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Summary

A shock converts ordered flow energy into heat, magnetic energy, and sometimes a population of energetic particles. We begin with the fluid conservation laws and show why a compressive disturbance steepens. Integrating those laws across a thin transition gives the shock jump conditions: a supersonic gas is compressed, slowed, and heated. Magnetic fields modify the fluxes, while collisionless plasma processes determine how the transition actually occurs.

The second half asks how a small fraction of the particles can gain much more than the thermal energy. Following the source notes, we first examine confinement and the evolution of a supernova blast wave, then compare second-order Fermi acceleration with repeated crossings of a shock. The energy gain per cycle and the probability of escaping together determine a power law. For a strong, nonrelativistic, unmodified shock, relativistic particles have the familiar spectrum $\mathrm{d}N/\mathrm{d}E \propto E^{-2}$.

Magnetic quantities use Gaussian-cgs units. Bulk shock speeds are nonrelativistic; accelerated particles may be relativistic.

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1 Start with the quantities a shock must conserve

Shocks occur when a flow cannot adjust smoothly through compressive waves: examples include Earth's bow shock, fast solar-wind disturbances, and supernova blast waves. A shock is a thin transition between two macroscopic states. Its internal structure may be complicated, but mass, momentum, and energy still have to balance across it.

We first consider an unmagnetized ideal gas with isotropic pressure, no external force, and no radiative or conductive energy loss. Let ρ, $\boldsymbol{u}$, and P denote mass density, bulk velocity, and pressure. The specific internal energy ϵ is energy per unit mass, and

$$\frac{D}{Dt} = \partial_t + \boldsymbol{u}\cdot\boldsymbol{\nabla}, \qquad P = (\gamma-1)\rho\epsilon, \qquad c_s^2 = \left.\frac{\partial P}{\partial \rho}\right|_{s_g} = \frac{\gamma P}{\rho}. \tag{1.1}$$

Here γ is the adiabatic index and s_g is the specific gas entropy. For a monatomic, nonrelativistic gas, $\gamma = 5/3$.

The familiar equations, away from the shock layer

The smooth, inviscid flow obeys

$$\frac{D}{Dt}\rho = -\rho\boldsymbol{\nabla}\cdot\boldsymbol{u}, \qquad \rho\frac{D}{Dt}\boldsymbol{u} = -\boldsymbol{\nabla}P, \qquad \rho\frac{D}{Dt}\epsilon = -P\boldsymbol{\nabla}\cdot\boldsymbol{u}. \tag{1.2}$$

The last equation describes compressional heating or expansion cooling. Together with the equation of state it gives $D(P/\rho^\gamma)/Dt = 0$ for a smooth adiabatic flow. We must not impose this constant-entropy relation *across* a dissipative shock.

Write each equation as a conservation law

A conservation law has the form $\partial_t Q + \boldsymbol{\nabla}\cdot\boldsymbol{F}_Q = 0$: the amount in a volume changes because a flux crosses its boundary. Using continuity, the equations become

$$\partial_t\rho + \boldsymbol{\nabla}\cdot(\rho\boldsymbol{u}) = 0, \tag{1.3}$$

$$\partial_t(\rho u_i) + \partial_j(\rho u_i u_j + P\delta_{ij}) = 0, \tag{1.4}$$

$$\partial_t\left[\rho\left(\frac{u^2}{2}+\epsilon\right)\right] + \boldsymbol{\nabla}\cdot\left[\rho\boldsymbol{u}\left(\frac{u^2}{2}+h\right)\right] = 0, \tag{1.5}$$

where repeated Cartesian indices are summed and

$$h = \epsilon + \frac{P}{\rho} = \frac{\gamma}{\gamma-1}\frac{P}{\rho} \tag{1.6}$$

is the specific enthalpy. The energy *density* contains ϵ; the energy *flux* contains h because pressure does work as matter crosses the boundary. The momentum flux similarly contains both advected momentum, $\rho u_i u_j$, and pressure, $P\delta_{ij}$.

> Conservation determines how the states on the two sides are related. Dissipation determines the transition between them and selects the entropy-increasing solution.

2 Why a compressive disturbance steepens

In linear acoustics, the perturbation is small and sound propagates at the background c_s. At finite amplitude, the flow itself helps carry the disturbance. Faster material behind slower material catches up with it, sharpening the gradient.

The simplest model: follow a characteristic

The one-dimensional momentum equation is

$$\partial_t u + u\partial_x u = -\frac{1}{\rho}\partial_x P. \tag{2.1}$$

Following the source notes, isolate nonlinear advection by neglecting the pressure force. This is a useful cold-flow illustration, not an equation for the completed shock:

$$\boxed{\partial_t u + u\partial_x u = 0.} \tag{2.2}$$

Along a curve satisfying $\mathrm{d}x/\mathrm{d}t = u$, the chain rule gives

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \partial_t u + \frac{\mathrm{d}x}{\mathrm{d}t}\partial_x u = 0. \tag{2.3}$$

Thus each characteristic carries a constant velocity. If its starting position is a and the initial velocity is $u_0(a)$,

$$\boxed{u = u_0(a), \qquad x = a + u_0(a)t.} \tag{2.4}$$

Take two nearby starting points separated by $\delta a > 0$. Their later separation is

$$\delta x = \left[1 + t\,u_0'(a)\right]\delta a, \qquad \partial_x u = \frac{u_0'(a)}{1 + t\,u_0'(a)}. \tag{2.5}$$

Where $u_0' < 0$, the rear element is faster than the front element. Their separation shrinks and the slope becomes increasingly negative. For a smooth initial profile, the first divergence occurs at

$$t_{\mathrm{steep}} = -\frac{1}{\min_a u_0'(a)}, \qquad \min_a u_0'(a) < 0. \tag{2.6}$$

An expanding region with $u_0' > 0$ instead spreads out. Compression, rather than finite amplitude alone, is the essential ingredient.

What replaces the mathematical singularity?

After characteristics intersect, this smooth-flow construction would assign several velocities to the same position. A single fluid velocity cannot do that. Pressure and small-scale dissipative or kinetic processes become important; they produce a narrow shock transition instead.

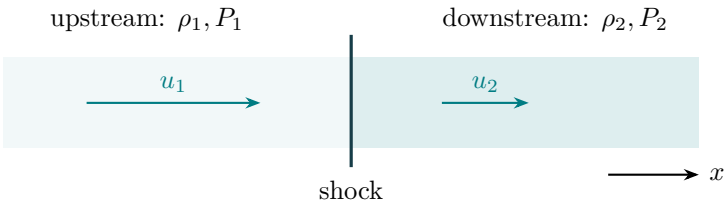
Pressure does not remove the basic steepening mechanism. In an ideal gas, sound characteristics move at $u \pm c_s$, and both u and c_s vary within a finite-amplitude compressive wave. Even a weak compressive wave can eventually steepen; its velocity amplitude need not initially exceed c_s.

> Nonlinear propagation brings neighboring characteristics together. The resulting small spatial scale activates the physics omitted from the smooth ideal equations. We can determine the jump without first resolving that small scale.

3 Integrate across the shock

Choose the shock frame once

Consider a stationary, planar shock at $x = 0$. Region 1 is upstream and region 2 is downstream. Choose $+x$ along the mass flow, so the normal speeds are positive: $u_1 > u_2 > 0$ for a compressive shock. All speeds in the jump conditions are measured in this shock frame.



For a steady one-dimensional conservation law, $\partial_x F_Q = 0$. Integrating through the layer gives $[F_Q] = 0$, where $[A] \equiv A_2 - A_1$. The mass, normal momentum, and energy fluxes therefore satisfy

$$j \equiv \rho_1 u_1 = \rho_2 u_2, \tag{3.1}$$

$$P_1 + \rho_1 u_1^2 = P_2 + \rho_2 u_2^2, \tag{3.2}$$

$$\frac{u_1^2}{2} + h_1 = \frac{u_2^2}{2} + h_2. \tag{3.3}$$

The last equation follows by dividing the conserved energy flux by $j \neq 0$. Tangential momentum conservation gives $j[\boldsymbol{u}_t] = 0$; in hydrodynamics a common tangential velocity can be removed by a change of frame.

Reduce the three equations to one compression ratio

Define

$$r \equiv \frac{\rho_2}{\rho_1} = \frac{u_1}{u_2}, \qquad M_1 \equiv \frac{u_1}{c_{s1}}, \qquad c_{s1}^2 = \frac{\gamma P_1}{\rho_1}. \tag{3.4}$$

Mass conservation gives $u_2 = u_1/r$. Momentum conservation then gives

$$P_2 = P_1 + \rho_1 u_1^2 \left(1 - \frac{1}{r}\right). \tag{3.5}$$

Insert these expressions and (1.6) into the energy equation, then divide by u_1^2:

$$\frac{1}{2} + \frac{1}{(\gamma-1)M_1^2} = \frac{1}{2r^2} + \frac{1}{(\gamma-1)M_1^2 r} + \frac{\gamma}{\gamma-1}\frac{1-1/r}{r}. \tag{3.6}$$

There is a trivial solution $r = 1$, representing no jump. The shock solution is

$$\boxed{r = \frac{(\gamma+1)M_1^2}{(\gamma-1)M_1^2 + 2}.} \tag{3.7}$$

For comparison, a surface with $j = 0$ is a contact or tangential discontinuity: pressure is continuous, but density and possibly tangential velocity can jump. No material crosses that surface.

4 What changes across a gas shock?

The compression ratio determines the remaining jumps. Substitution into (3.5) gives

$$\boxed{\frac{P_2}{P_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1}, \qquad \frac{T_2}{T_1} = \frac{P_2/P_1}{r}.} \tag{4.1}$$

The temperature ratio assumes the same composition and a single gas temperature, $P = \rho k_B T/\bar{m}$, where $\bar{m}$ is the mean mass per thermal particle.

The strong-shock limit

For $M_1 \gg 1$, the upstream thermal pressure is negligible compared with the ram pressure:

$$r \longrightarrow \frac{\gamma + 1}{\gamma - 1}, \qquad P_2 \simeq \frac{2}{\gamma + 1}\rho_1 u_1^2, \qquad k_B T_2 \simeq \frac{2(\gamma - 1)}{(\gamma + 1)^2}\bar{m}u_1^2. \tag{4.2}$$

For $\gamma = 5/3$, these become

$$\boxed{\rho_2 = 4\rho_1, \qquad u_2 = \frac{u_1}{4}, \qquad P_2 \simeq \frac{3}{4}\rho_1 u_1^2, \qquad k_B T_2 \simeq \frac{3}{16}\bar{m}u_1^2.} \tag{4.3}$$

The downstream bulk speed is smaller and the random energy is larger. A finite compression limit results because the heated gas pushes back. The factor four is a result for a nonradiative, ideal monatomic gas; it is not a universal maximum for every shock.

Supersonic in, subsonic out

Using $c_{s2}^2 = \gamma P_2/\rho_2$, the downstream Mach number is

$$M_2^2 = \frac{1 + (\gamma - 1)M_1^2/2}{\gamma M_1^2 - (\gamma - 1)/2}. \tag{4.4}$$

For $M_1 > 1$, we obtain $M_2 < 1$. Upstream acoustic signals cannot run ahead of the incoming flow in the shock frame; downstream acoustic communication is possible. In the strong-shock limit, $M_2^2 \to (\gamma - 1)/(2\gamma)$, or $1/5$ for $\gamma = 5/3$.

Why the entropy condition matters

Conservation alone supplies algebraic jump conditions. The physical solution must also increase entropy:

$$s_{g2} - s_{g1} = c_v \ln\left[\frac{P_2/P_1}{r^\gamma}\right] > 0, \qquad c_v = \frac{k_B}{(\gamma - 1)\bar{m}}. \tag{4.5}$$

The compressive branch with $M_1 > 1$ satisfies this condition. The corresponding formal expansion-shock branch has decreasing entropy and is inadmissible; a physical expansion is resolved by a rarefaction instead.

> A nonradiative shock conserves total energy while producing entropy. "No energy radiated away" does not mean "reversible" or "constant entropy." The lost bulk kinetic energy becomes internal energy through processes within the transition.

5 Add a magnetic field: the MHD jump conditions

Magnetic pressure, tension, and Poynting flux now participate in the balance. For ideal MHD with isotropic gas pressure, the momentum-flux tensor is

$$\Pi_{ij} = \rho u_i u_j + \left(P + \frac{B^2}{8\pi}\right)\delta_{ij} - \frac{B_i B_j}{4\pi}. \tag{5.1}$$

Let n denote the shock-normal direction and t the tangential directions. A steady planar shock obeys

$$[\rho u_n] = 0, \qquad [B_n] = 0, \tag{5.2}$$

$$[u_n \boldsymbol{B}_t - B_n \boldsymbol{u}_t] = 0, \tag{5.3}$$

$$\left[\rho u_n \boldsymbol{u}_t - \frac{B_n \boldsymbol{B}_t}{4\pi}\right] = 0, \tag{5.4}$$

$$\left[\rho u_n^2 + P + \frac{B_t^2 - B_n^2}{8\pi}\right] = 0, \tag{5.5}$$

$$\left[\rho u_n \left(\frac{u^2}{2} + h\right) + \frac{u_n B^2 - B_n(\boldsymbol{u}\cdot\boldsymbol{B})}{4\pi}\right] = 0. \tag{5.6}$$

The condition $[B_n] = 0$ follows from $\boldsymbol{\nabla}\cdot\boldsymbol{B} = 0$. Continuity of tangential electric field, with $\boldsymbol{E} = -\boldsymbol{u}\times\boldsymbol{B}/c$, gives (5.3). The remaining conditions conserve tangential momentum, normal momentum, and total energy. The magnetic term in (5.6) is the normal Poynting flux.

Two simple geometries

Parallel shock: field normal to the front. On the branch with $\boldsymbol{B}_t = \boldsymbol{u}_t = 0$ on both sides, B_n is unchanged and its normal stress cancels from the jump. The Poynting flux vanishes. The gas jump conditions are therefore the hydrodynamic ones. This specifies a branch; an upstream parallel field alone does not exclude other MHD solutions.

Perpendicular shock: field tangent to the front. Set $B_n = 0$ and choose $\boldsymbol{u}_t = 0$. Combining $[u_n \boldsymbol{B}_t] = 0$ with mass conservation gives

$$\boxed{\frac{B_{t2}}{B_{t1}} = \frac{\rho_2}{\rho_1} = r.} \tag{5.7}$$

The remaining normal balances reduce to

$$\left[\rho u_n^2 + P + \frac{B_t^2}{8\pi}\right] = 0, \qquad \left[\frac{u_n^2}{2} + h + \frac{B_t^2}{4\pi\rho}\right] = 0. \tag{5.8}$$

Compression strengthens the transverse field; the magnetic pressure resists further compression and some incoming energy goes into the field.

Which signal speed is relevant?

For a *fast* MHD shock, the upstream normal speed exceeds the fast magnetosonic speed

$$c_f^2 = \frac{1}{2}\left[c_s^2 + v_A^2 + \sqrt{(c_s^2 + v_A^2)^2 - 4c_s^2 v_A^2 \cos^2\theta}\right], \qquad v_A^2 = \frac{B^2}{4\pi\rho}, \tag{5.9}$$

where θ is the angle between the field and the shock normal. Slow shocks are a different MHD family. Thus "super-fast" is not a condition on every MHD shock. When gas pressure greatly exceeds magnetic pressure, $c_f \simeq c_s$ and the fast-shock condition approaches the gas result.

6 What makes a collisionless shock possible?

The jump conditions tell us the net changes, but not the shock thickness or the way particles acquire their downstream distribution. Those questions require the microscopic processes inside the layer.

Collisions can regularize a steep gradient

For an ordinary collisional fluid, the viscous force becomes important as the velocity gradient steepens. A schematic balance is

$$u\frac{\Delta u}{\Delta} \sim \nu\frac{\Delta u}{\Delta^2} \quad \Longrightarrow \quad \Delta \sim \frac{\nu}{u}, \tag{6.1}$$

where Δ is the transition thickness and ν the kinematic viscosity. Since $\nu \sim v_{\rm th}\lambda_{\rm mfp}$, a strong gas shock has a thickness of order a few mean free paths, up to kinetic factors. In a neutral gas the estimate $\lambda_{\rm mfp} \sim 1/(n\sigma)$ uses the atomic collision cross section.

In a dilute ionized plasma, the relevant binary relaxation is Coulomb scattering. For a fast supernova-remnant shock, its relaxation length can exceed the remnant size. A nuclear hard-sphere cross section is not the appropriate estimate of that Coulomb mean free path. Yet sharp shocks are observed: binary collisions cannot be what sustains the transition.

Collective electromagnetic fields replace binary relaxation

Particles are deflected and sometimes reflected by electromagnetic fields. The resulting beams and anisotropies drive fluctuations that scatter the particles and redistribute flow energy. Self-generated fields can therefore mediate a shock even when Coulomb collisions are negligible.

Relevant ion scales include

$$r_{gi} = \frac{u_1}{\Omega_{ci}} = \frac{m_i c u_1}{ZeB}, \qquad d_i = \frac{c}{\omega_{pi}}, \qquad \Omega_{ci} = \frac{ZeB}{m_i c}, \quad \omega_{pi}^2 = \frac{4\pi n_i Z^2 e^2}{m_i}. \tag{6.2}$$

For protons,

$$r_{gi} \simeq 1.0 \times 10^{11}\,\mathrm{cm}\left(\frac{u_1}{10^4\,\mathrm{km\,s^{-1}}}\right)\left(\frac{B}{1\,\mu\mathrm{G}}\right)^{-1}. \tag{6.3}$$

This is much smaller than a parsec (3.09×10^{18} cm). The relevant shock scales depend on magnetization, obliquity, and the instabilities present; the entire transition is not always one gyroradius thick.

Conservation does not fix the energy partition

Without rapid collisional equilibration, electrons and ions need not have the same temperature. A nearly thermal core may also coexist with a nonthermal tail. The total downstream pressure and energy can satisfy the jump conditions even when $T_e \neq T_i$ and neither species is exactly Maxwellian.

> "Collisionless" does not mean "no interaction." Collective fields exchange momentum and energy with the particles. Fluid conservation remains useful far from the layer, while kinetic physics determines heating, reflection, and injection into acceleration.

We now follow the particles that return to the shock repeatedly rather than simply joining the downstream thermal population.

7 Cosmic rays: can a source contain the particles?

Cosmic rays are energetic particles of extraterrestrial origin, predominantly protons and nuclei. Their spectrum extends to energies of order $10^{20}\,\mathrm{eV}$ and above. A useful scale is $10^{20}\,\mathrm{eV} \simeq 16\,\mathrm{J}$: a macroscopic amount of energy carried by one particle.

Supernova-remnant shocks are leading candidates for accelerating Galactic cosmic rays, particularly below the spectral knee at a few $10^{15}\,\mathrm{eV}$. This motivates two separate questions: can a remnant confine particles at those energies, and what spectrum does shock acceleration produce?

The Hillas confinement estimate

For an ultrarelativistic particle, $E \simeq pc$ and a characteristic gyroradius is

$$r_L \sim \frac{pc}{ZeB} \simeq \frac{E}{ZeB}, \qquad r_L \lesssim L \quad \Longrightarrow \quad \boxed{E \lesssim ZeBL.} \tag{7.1}$$

This is a necessary size estimate for magnetic confinement. Numerically,

$$r_L \simeq 1.08\,\mathrm{pc}\,\frac{E/(1\,\mathrm{PeV})}{Z\,[B/(1\,\mu\mathrm{G})]}. \tag{7.2}$$

A 3 PeV proton in a $1\,\mu$G field has $r_L \simeq 3.2\,\mathrm{pc}$, already larger than a remnant of radius 1 pc. Simple compression amplifies a perpendicular field only by r, approaching four in the strong hydrodynamic limit. Additional amplification can be important: the current carried by accelerated particles can drive upstream magnetic instabilities, including the Bell instability [1].

The source evolves: a supernova blast wave

Consider a spherical, nonradiative explosion of conserved energy E_{SN} into a uniform medium of density ρ_0. Once the swept-up mass greatly exceeds the ejecta mass, dimensional analysis gives the Sedov–Taylor scaling:

$$\boxed{R(t) = \xi \left(\frac{E_{\mathrm{SN}} t^2}{\rho_0}\right)^{1/5}, \qquad u_{\mathrm{sh}} = \frac{\mathrm{d}R}{\mathrm{d}t} = \frac{2R}{5t}.} \tag{7.3}$$

Here ξ is a dimensionless number fixed by the gas equation of state. One can see the scaling directly from $E_{\mathrm{SN}} \sim \rho_0 R^3 u_{\mathrm{sh}}^2$: more swept-up mass means a slower shock at fixed explosion energy. The assumptions are

$$\frac{4\pi}{3}\rho_0 R^3 \gg M_{\mathrm{ej}}, \qquad \text{radiative and escaping-particle energy losses are small.} \tag{7.4}$$

This stage is neither early free expansion nor the later strongly radiative stage.

> The Hillas condition tests whether an orbit fits. It does not establish that acceleration is fast enough. A successful accelerator must also reach the required energy before the source evolves, particles escape, or losses remove their energy.

8 Fermi's idea: gain energy from moving scatterers

Magnetic forces do no work in a frame where the scattering field is stationary. But a particle scattered by a moving magnetic structure can change energy in the observer's frame. The energy comes from the motion of that structure, through the associated electric field.

One encounter with a moving magnetic mirror

Use the source's simple reflection model. A magnetic mirror moves at velocity $-V\hat{\boldsymbol{x}}$, with $V \ll c$. An incoming ultrarelativistic particle has $p_x \simeq (E/c)\mu$, where $\mu = \cos\theta$. Positive μ corresponds to a head-on encounter. With $b = V/c$ and $\Gamma_V = (1-b^2)^{-1/2}$, transforming to the mirror frame gives

$$E' = \Gamma_V(E + Vp_x), \qquad p'_x = \Gamma_V\left(p_x + \frac{VE}{c^2}\right). \tag{8.1}$$

Reflection is elastic in that frame: $E'_{\rm out} = E'$ and $p'_{x,\rm out} = -p'_x$. Transforming back,

$$E_{\rm out} = \Gamma_V(E' + Vp'_x) = \Gamma_V^2 E(1 + 2b\mu + b^2). \tag{8.2}$$

Therefore, to second order,

$$\boxed{\frac{\Delta E}{E} = 2b\mu + 2b^2 + O(b^3).} \tag{8.3}$$

The first-order term is positive for head-on encounters and negative for overtaking encounters. A single encounter need not increase the energy.

Random encounters: second-order Fermi acceleration

Suppose the particles are isotropic before encountering an ensemble of randomly moving mirrors. Head-on encounters are slightly more frequent because their relative approach speed is greater. For this model, the normalized encounter weighting is

$$w(\mu)\,\mathrm{d}\mu = \frac{1}{2}(1 + b\mu)\,\mathrm{d}\mu, \qquad -1 \le \mu \le 1. \tag{8.4}$$

Thus $\langle\mu\rangle = b/3$ and

$$\boxed{\left\langle \frac{\Delta E}{E} \right\rangle = 2b\,\langle\mu\rangle + 2b^2 = \frac{8}{3}\left(\frac{V}{c}\right)^2.} \tag{8.5}$$

The coefficient $8/3$ belongs to this reflection prescription; a different scattering model can change it. The robust point is the *second-order* scaling: head-on gains and overtaking losses cancel at first order. Random moving magnetic fluctuations in turbulence provide the physical setting for this kind of acceleration.

To obtain a first-order gain, we need a systematic convergence of scattering centers. A shock supplies it: the upstream and downstream fluids approach one another in their relative rest frames.

9 A shock supplies first-order Fermi acceleration

Return to the shock frame of Section 3. Both flows move in the same $+x$ direction, but $u_1 > u_2$. Their relative speed is

$$U \equiv u_1 - u_2 = u_1\left(1 - \frac{1}{r}\right) > 0. \tag{9.1}$$

In the upstream rest frame, the downstream fluid moves toward the upstream at $-U$; in the downstream rest frame, the upstream moves toward the downstream at $+U$. A particle that crosses and returns samples converging scattering centers on both legs.

Assumptions of the basic DSA model

Take a steady, nonrelativistic shock, $u_1, u_2 \ll c$, and relativistic energetic particles with speed $v \simeq c$. Scattering approximately isotropizes them in each local fluid frame and is elastic there. The scatterers move with the fluid, and the energetic particles are a *test population*: their pressure does not modify the shock. We also neglect radiative losses during one cycle.

Repeated scattering lets particles cross the shock many times. This is *diffusive shock acceleration* (DSA). The direction distribution is nearly isotropic within each region, but particles crossing a plane are not an unbiased sample of that distribution.

Average over particles that actually cross

For a crossing, let $\mu \in [0, 1]$ be the cosine of the angle to the normal directed toward the other region. The crossing flux includes a factor $c\mu$. Its normalized angular distribution and mean are

$$w_{\rm cross}(\mu) = 2\mu, \qquad \langle\mu\rangle_{\rm cross} = \int_0^1 2\mu^2\,\mathrm{d}\mu = \frac{2}{3}. \tag{9.2}$$

The first-order change of energy on transforming into the approaching fluid frame is $\Delta E/E \simeq (U/c)\mu$. Hence each crossing contributes a mean gain $2U/(3c)$. For a complete upstream–downstream–upstream cycle, measured in the same fluid frame at the beginning and end,

$$\boxed{g \equiv \left\langle \frac{\Delta E}{E} \right\rangle_{\rm cycle} = \frac{4}{3}\frac{u_1 - u_2}{c}.} \tag{9.3}$$

Unlike random Fermi encounters, both legs have the same sign. The gain is first order in the flow speed divided by c.

The magnetic irregularities turn particles around; they are not a free energy source. In the shock frame their motion produces electric fields that do work. Ultimately, the accelerated particles draw on the incoming flow's bulk kinetic energy.

Why the energy gain alone is not the answer

Some particles are advected far downstream and never return. To find a spectrum we must combine the gain of the survivors with the probability of losing particles on every cycle.

> The slope is set by a competition: repeated crossings multiply particle energies, while downstream escape reduces the number that reach those energies. We now calculate both factors in the same approximation.

10 Gain plus escape gives a power law

Estimate the downstream escape probability

Let $n_{\rm cr}$ be the density of the nearly isotropic energetic-particle population just downstream. To leading order in u_2/c, the flux of particles crossing a plane toward the shock is

$$J_{\rm cross} = \int_{\mu>0} \frac{n_{\rm cr} c\mu}{4\pi}\,\mathrm{d}\Omega = \frac{n_{\rm cr} c}{4}. \tag{10.1}$$

The net flux carried away downstream is $J_{\rm adv} = n_{\rm cr} u_2$. Dividing it by the leading crossing flux gives the probability of escape per cycle,

$$P_{\rm esc} \simeq \frac{J_{\rm adv}}{J_{\rm cross}} = \frac{4u_2}{c}, \qquad \mathcal{P} \equiv 1 - P_{\rm esc} \simeq 1 - \frac{4u_2}{c}. \tag{10.2}$$

Here $\mathcal{P}$ is the retention probability. The estimate assumes no separate upstream escape boundary and an approximately uniform downstream distribution.

Eliminate the number of cycles

Write the mean energy multiplication per cycle as $A = 1 + g$. After k cycles,

$$E_k = E_0 A^k, \qquad N_k = N_0 \mathcal{P}^k. \tag{10.3}$$

If A and $\mathcal{P}$ are approximately independent of energy, eliminating k gives the cumulative distribution

$$N(>E) \propto \left(\frac{E}{E_0}\right)^{\ln \mathcal{P}/\ln A}. \tag{10.4}$$

The positive differential spectrum is $\mathrm{d}N/\mathrm{d}E \equiv -\mathrm{d}N(>E)/\mathrm{d}E$, so

$$\frac{\mathrm{d}N}{\mathrm{d}E} \propto E^{-s}, \qquad s = 1 - \frac{\ln \mathcal{P}}{\ln A} \simeq 1 + \frac{P_{\rm esc}}{g} = 1 + \frac{3u_2}{u_1 - u_2} = \boxed{\frac{r+2}{r-1}}. \tag{10.5}$$

For a strong gas shock with $\gamma = 5/3$, Section 4 gave $r = 4$. Therefore

$$\boxed{\frac{\mathrm{d}N}{\mathrm{d}E} \propto E^{-2}.} \tag{10.6}$$

The compression measured by the fluid jump conditions has determined the spectral slope of the accelerated test particles. An E^{-2} spectrum contains equal particle energy per logarithmic interval, since $E^2\mathrm{d}N/\mathrm{d}E$ is constant.

This result describes relativistic particles accelerated at a nonrelativistic shock. It is not a universal spectrum for all shocks, nor the spectrum measured at Earth after propagation. Injection, finite age, losses, particle back reaction, and scattering-center drift can change the outcome.

11 The physical chain and its limits

Step	What the model tells us
Nonlinear steepening	Compressive characteristics converge; a narrow transition forms.
Flux conservation	Mass, momentum, and energy connect the two sides, independently of the detailed shock thickness.
Gas shock	For a strong monatomic shock, $r = 4$ and $k_B T_2 = 3\bar{m}u_1^2/16$.
Magnetic field	Maxwell stress and Poynting flux modify the jumps; perpendicular compression gives $B_{t2} = rB_{t1}$.
Collisionless transition	Collective fields mediate reflection, scattering, and heating; species need not equilibrate.
Repeated crossings	Converging scattering centers produce a first-order mean gain $g = 4(u_1 - u_2)/(3c)$.
Downstream escape	$P_{\rm esc} = 4u_2/c$ competes with the gain, giving $s = (r+2)/(r-1)$ and $s = 2$ for $r = 4$.

Keep three questions separate

The jump: how much flow energy must be redistributed? The conservation laws answer this once the pressure model, magnetic geometry, and energy losses are specified.

The microscopic transition: which particles are heated, reflected, or injected? A fluid jump condition does not determine the electron–ion energy partition or the fraction of particles that enter DSA.

The accelerator: how far and how fast can the injected particles climb in energy? The cycle argument gives a slope. Confinement, scattering rates, source age, and losses determine the maximum energy and efficiency.

Do not confuse two spectral conventions

For an isotropic phase-space density $f(p)$, the momentum spectrum is $\mathrm{d}N/\mathrm{d}p = 4\pi p^2 f(p)$, apart from the common volume factor. The test-particle DSA result can also be written

$$f(p) \propto p^{-q}, \qquad q = \frac{3r}{r-1}, \qquad \frac{\mathrm{d}N}{\mathrm{d}E} \propto E^{2-q} \quad (E \simeq pc). \tag{11.1}$$

Thus $r = 4$ gives $f(p) \propto p^{-4}$ and $\mathrm{d}N/\mathrm{d}E \propto E^{-2}$: these are the same result expressed for different quantities. The conversion changes when particles are nonrelativistic.

Source and further reading

Primary source for this chapter. L. Sironi, *NSF GPAP 2026 lecture notes on shocks and particle acceleration*, supplied as `NSFGPAP2026_Sironi1.pdf`, PDF pp. 1–14. The sequence and principal derivations here follow those handwritten notes; the slides appended from PDF p. 15 onward are not included.

References

[1] A. R. Bell (2004), Turbulent amplification of magnetic field and diffusive shock acceleration of cosmic rays, *MNRAS* **353**, 550–558. The upstream current-driven amplification mechanism.

[2] R. Fitzpatrick, MHD Shocks. A complementary derivation of the magnetic jump conditions and shock families.